Canto-L

Training sample study · validation set · track builder: MWS-layer + learned merging (LM), Canto-L's frozen builder

	Canto-L (HNL) 1 seed · preview · HNL-trained	Canto-L (tt̄+HNL) 1 seed · preview · mixed	Canto-L (tt̄) 3 seeds · ttbar-trained	acorn (HNL) 3 seeds · HNL-trained	acorn (tt̄+HNL) 3 seeds · mixed	acorn (tt̄) 3 seeds · ttbar-trained
Displaced tracks |d0| ≥ 10 mm or r_v ≥ 100 mm	**99.2%**	**99.2%**	99.1%	98.6%	97.9%	97.2%
Mid-displaced |d0| 10–30 mm	99.0%	99.4%	**99.7%**	98.8%	98.7%	98.0%
Displaced |d0| 30–100 mm	**99.4%**	**99.4%**	99.3%	99.2%	98.4%	97.8%
Far-displaced |d0| 100–300 mm	**98.8%**	**98.8%**	97.6%	97.2%	95.6%	93.8%
Very far-displaced |d0| ≥ 300 mm	**94.7%** 18/19 particles	**94.7%** 18/19 particles	87.7% 50/57 particles	91.2% 52/57 particles	84.2% 48/57 particles	77.2% 44/57 particles
Prompt muons W → μ in HNL events	**99.8%**	**99.8%**	**99.8%**	**99.8%**	**99.8%**	**99.8%**
Dense prompt tracks ttbar, |d0| < 1 mm	**96.7%**	96.4%	96.3%	94.9%	95.2%	95.5%
Fake-track rate lower is better	1.02%	**1.01%**	1.03%	1.37%	1.39%	1.53%
Track builder same for every model	MWS + LM	MWS + LM	MWS + LM	MWS + LM	MWS + LM	MWS + LM
Training data	HNL signal	tt̄ + HNL	tt̄ only	HNL signal	tt̄ + HNL	tt̄ only
Parameters edge classifier	348k	348k	348k	264k	264k	264k

All numbers: ACTS generic detector simulation without pileup, validation set only (2,999 events of ttbar and the two HNL samples: the displaced-track, prompt-muon and fake-rate rows from the HNL events, 'Dense prompt tracks' from the ttbar events). Track builder for every model: MWS-layer + learned merging (LM), Canto-L's frozen builder, its setting chosen per seed on validation by the builder rule at track fake rate ≤ 2.94 % (FR*). Track efficiency by double-majority matching; mean over the listed training seeds. Best value per row bold and shaded (ties included).

Canto-L (HNL) and Canto-L (tt̄ + HNL), previews: the frozen Canto-L recipe (learned triplet-support union, edge classifier, 30 epochs) with only the training sample changed (HNL: 15,999 graphs of both HNL samples; tt̄ + HNL: 7,999 + 7,999), one seed each; Canto-L (tt̄): the frozen Canto-L, three seeds. acorn: acorn's interaction network trained on the same three samples, three seeds each.

Particles per seed: displaced 1,331; |d0| 10–30 mm 508, 30–100 mm 502, 100–300 mm 167, ≥ 300 mm 19; W muons 1,869; tt̄ prompt 52,702. One seed resolves differences of about 0.5 point in the displaced row.